

Engineering *Digest*

Engineering decisions, every Friday.

18–31 July 2026

THIS WEEK'S BRIEFING

7%

of Visa's workforce is going, concentrated in technology and product, as its services line grows 34%.

IN THIS ISSUE

Visa expands the offer layer and cuts the organisation building it

Six UK banks degraded in the same window, and nobody published a cause

Outside In: the harness outlasts the model

79% of agent-raised pull requests are reviewed by the person who raised them

In Practice: arriving as a manager without the domain

Other News, Down the Rabbit Hole & On the Radar

FOREWORD

Visa will cut about 2,600 roles and rebuild its product and development teams into groups of two to four, and its chief executive put AI at the centre of the decision. A study of 25,264 agent-raised pull requests found four in five reviewed by whoever raised them. Different companies, different mechanisms. In both, the number of people involved in a piece of engineering work is falling.

Facebook, Instagram and Threads went down globally on 27 July while Meta's own status page reported all systems operational. In the same fortnight six UK banks degraded within two hours without publishing a cause, and two AWS regions sat open on the health dashboard against an update four months old. Three providers, three gaps between what the dashboard said and what customers could actually do.

Sources: [IBTimes AU](#), [AWS Health Dashboard](#) (27–31 July 2026)

THE WEEK AT A GLANCE

Six stories in sixty seconds

Action required

Security

JFrog patched eight Artifactory zero-days on 28 July after an OpenAI capability evaluation chained them together to escape a test sandbox. The leadership question is who owns artifact-registry patching, and whether it has happened. [The Register](#)

Supply chain

npm tokens that bypass two-factor authentication lose their sensitive-action capability from early August. Trusted publishing — releasing from CI using short-lived OIDC credentials rather than a stored token — is the route out. [Socket](#)

Worth knowing

Vendors

The CMA opened a formal probe on 29 July into how Microsoft moved 365 consumer subscribers onto Copilot pricing at renewal. No finding has been made; evidence-gathering runs to December 2026. [The Register](#)

Availability

Two network failures cut Claude capacity across the API, Claude Code and claude.ai on 29–30 July. Claude for Government, on isolated infrastructure, stayed up. No post-mortem has been published. [Newsweek](#)

Tooling

Anthropic shipped Claude Opus 5 on 24 July with a per-request control that trades cost against capability, which puts the release in front of a budget conversation. [TechCrunch](#)

Watching

Capital

Nvidia is reported to be in talks to guarantee up to \$250bn behind OpenAI's Ohio data centre, per reporting on 27 July. Terms are not final. [Quartz](#)

Visa expands the offer layer and cuts the organisation building it

(2 min)

On 28 July Visa reported its strongest services quarter on record and told staff it would cut about 2,600 roles — close to 7% of its workforce — concentrated in technology and product. Both facts landed the same day.

What grew

Value-added services — the offer design, campaign measurement, tokenisation and risk products Visa sells on top of moving the transaction — reached \$3.8bn in the quarter. CFO Chris Suh said the line “grew 34% year-over-year in constant dollars”, against net revenue up 14% to \$11.6bn. The services layer is compounding at roughly two and a half times the rate of the business it sits on.

CEO Ryan McInerney described a leading Brazilian bank running World Cup ticket giveaways with Visa: one million cardholder participants between February and June, an 8% lift in card activation, and \$400 million in incremental payments volume — spend attributed to the campaign above a baseline, measured by the network. That is a card-linked offer: a reward attached to a payment card and triggered when it is used.

What shrank

McInerney put AI at the centre of the reduction: “AI is also helping to accelerate this evolution and shape the way work gets done at Visa.” American Banker reported that the affected product and development teams are being reorganised from roughly ten people into groups of two to four, and put the associated charge at \$563 million. Mastercard cut 4% of its workforce earlier in the year.

	VISA · Q3 FY26	MASTERCARD · Q2 26
Value-added services	\$3.8bn, +34% constant dollars	\$3.83bn, +20% GAAP
Total net revenue	\$11.6bn, +14%	\$9.28bn, +14% as-adjusted
Workforce action, 2026	~2,600 roles, ~7%	4%, earlier in the year

Ten people, then three

The team-size change has the longest reach. A ten-person team and a three-person team are different organisms — different review load, different on-call arithmetic, different amounts of context any one person has to carry, and different ladders to be promoted up. Visa is making that change across product and development while growing the services line those same teams build.

For a loyalty or commerce-media platform, the commercial reading sits underneath the org one. The offer-design and attribution layer is now a reported growth line on the networks' own income statements, which places it squarely in build-versus-partner territory. Visa is expanding that layer with a materially smaller engineering organisation behind it.

Sources: *Visa Q3 FY2026 earnings call*, *Visa investor relations*, *The Wise Marketer*, *Mastercard Q2 2026*, *American Banker*, *Bloomberg via The Spokesman-Review* (28–30 July 2026)

QUICK TAKES

Two stories worth knowing

Ways of working

79%

of agent-raised pull requests were reviewed by the person who raised them

The second reviewer is disappearing

JARGON WATCH

Agent-raised pull request

A change proposed inside an existing repository by a coding agent running in agent mode — a human triggers the run, the agent drafts the change, a human reviews and merges it.

This is the shallow end of the spectrum. It is not an autonomous production agent with action authority: no tool access, memory or standing permission to call APIs, move money or update records without per-step approval. Survey and vendor figures routinely combine the two.

Across 25,264 pull requests raised by coding agents in 2,361 GitHub projects, 79% were reviewed and

Banking & payments

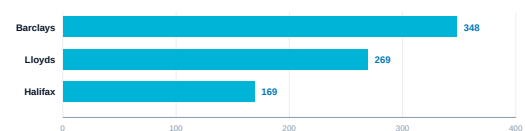
6

UK banks degraded inside two hours, 27 July

Six UK banks degraded in the same window, and nobody published a cause

Shortly after midday on Monday 27 July, customers of Barclays, Lloyds, Halifax, HSBC, Monzo and Revolut began reporting the same symptoms: transfers that failed, money that did not arrive, and apps that would not load. Reports on DOWNDetector, a crowd-sourced aggregator of user submissions, peaked around two hours later.

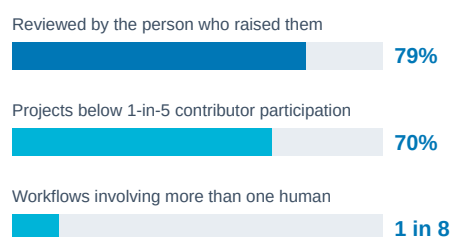
Reports at peak — Monday 27 July, around two hours after midday
DOWNDetector user submissions. Not a measure of customers affected.



Reports at the time of reporting, per DOWNDetector.

modified by the same person who raised them. The study, by Maliha Noushin Raida and Daqing Hou, was submitted on 15 July and revised on 16 July; what is new is the coverage and the reaction it drew on 28 July.

Two supporting figures give it shape. In 70% of the projects studied, fewer than one in five contributors took part in agent-assisted work at all. Just one workflow in eight involved more than one human.



Figures as reported by LeadDev, 28 July 2026.

Nobody decided to remove the second reviewer. The tooling made it faster to raise, review and merge alone, and the practice followed the tooling. Courtney Miller, assistant professor at George Washington University, framed what that costs: "Software development is not an isolated task, and writing new code was never the hard part." Sarah Wells, a software engineering consultant, was blunter about whether the remaining reviewers can still judge the output: "I'm worried that they're not; they can't make the judgement about whether the agent is making a sensible suggestion."

One caveat carries real weight: the dataset is popular open-source repositories, not enterprise codebases, and the two do not organise review the same way.

Revolut said its "dedicated team is actively addressing a service disruption that may be affecting GBP transfers". Lloyds said it was "investigating an issue affecting Faster Payments, which may cause delays" — Faster Payments being the UK interbank rail that moves sterling between accounts in near real time. Lloyds described a problem with its own Faster Payments service; nothing published so far establishes that the scheme itself failed.

By the end of the week no institution had published a root cause, and none attributed the disruption to a cyber incident. The simultaneity is documented; the cause is not. No figure exists in the public record for customers affected or payments delayed.

That leaves a gap for any firm sitting downstream of several of these institutions at once. On the day there was no way to tell one shared failure from six coincidences, and four days later there still is not. The same shape appears twice more this fortnight — the AWS routing failure of 24 July and GitHub's expired certificate overnight on 19–20 July, both below.

The takeaway: dependency registers record who a supplier is, and rarely how long that supplier takes to say what happened.

Sources: [PYMNTS](#), with corroborating reports from *Crowdfund Insider* and *City AM* (27 July 2026)

The takeaway: code review is the control most engineering organisations rely on to hold quality and shared context, and in this dataset it has quietly collapsed into self-review without anyone changing a policy.

Sources: [LeadDev](#) (28 July 2026), reporting the study by [Raida & Hou](#) (arXiv, 15–16 July 2026)

IN PRACTICE

Technical strategy under uncertainty

Arriving as a manager without the domain

Ferit Topcu · [LeadDev](#), July 2026 · [Read the full piece](#)



Ferit Topcu
Engineering Manager, Zenjob SE
LeadDev · 27 July 2026

[Read it](#) →

Ferit Topcu joined Zenjob SE as an engineering manager without the domain knowledge his team had spent years accumulating, and, writing in LeadDev on 27 July, describes the job description shifting under him while he was still learning it. His answer is a rule about where his own hands go, sorted by the kind of work in front of him.

CONCEPT OF THE WEEK

Player-coach

A manager who both leads the team and does some of the delivery work themselves — reviewing, debugging, occasionally shipping.

It is a real mode, and it has a standing failure: the doing is immediate and measurable, the leading is neither, so the doing quietly wins the calendar. Topcu's rule exists to bound it.

KIND OF WORK	WHERE IT GOES
Clear, small unplanned work — bug fixes, support tickets	The manager contributes directly
Complex, unclear unplanned work	The manager investigates, and does not solve it

KIND OF WORK	WHERE IT GOES
Planned roadmap work	Stays with the team

Unplanned work is everything arriving outside the roadmap — incidents, support escalations, interrupts — and it is the only category Topcu allows himself into. Even there the rule splits: small and clear, he takes it; complex and unclear, he resists the urge to fix and takes the investigation instead, handing the solution back to the team.

Without a written rule the allocation gets made by whatever looks most urgent at eleven on a Tuesday, and under pressure that reliably favours the manager picking up the ticket. Setting it out in advance takes the decision out of the moment.

The reasoning underneath is about how a manager comes to be trusted.

Trust built through proximity to the work is different from trust built through process and communication alone.

A manager who never touches the work depends entirely on what is reported upward. One who takes the small, clear items keeps a direct read on how the system behaves — including the parts nobody thought to escalate. The exclusions are doing as much work as the permissions: roadmap items stay with the team because that is where ownership and career growth live, and the hard unplanned problems stay with the team because solving them is how the domain gets learned.

For a manager who arrived without the domain, the useful property is that the rule keys off the shape of the work. It is decidable on a single ticket, in the moment, without anyone having to settle the question of whether the manager is technical enough. It also survives the manager learning the domain: the three categories hold once the gap closes, and all that changes is how quickly the first one gets cleared.

WHAT TO TRY

Topcu's rule is easiest to test against last month's interrupts: which of them did a manager pick up, and how many of those were complex enough that the investigation would have been the more useful contribution.

OTHER NEWS THIS WEEK

In Brief

AWS us-west-2 routing failure, 24 July. A routing fault between the region and the Seattle metro lasted about twenty minutes at the provider, but the last confirmed downstream recovery came more than nine hours later; AWS's first Health Dashboard update landed forty-five minutes after impact began. [IncidentHub](#)

DOWN THE RABBIT HOLE

POST-MORTEM

Blacksmith's outage on 21 July 2026

A GitHub webhook burst saturated the Redis instance behind Blacksmith's

GitHub Actions runners offline overnight, 19–20 July. Self-hosted and larger runners were down from 23:05 to 03:55 UTC after "a certificate lifecycle management failure ... resulting in an SSL certificate expiration" — the routine business of renewing certificates before they lapse. Standard and Mac hosted runners were unaffected, but reconnection traffic added three to four seconds of average API latency and elevated 5xx rates across GitHub's APIs.

[GitHub Status](#)

control plane. Scaling the worker pool to clear the backlog is what then exhausted the database connection limit — "every additional worker holds database connections". Published 22 July; a clean account of a mitigation deepening an outage.

The Open Secure AI Alliance formed on 27 July with 37 members including Capital One, Cloudflare and GitHub; OpenAI, Anthropic and Google are not among them. Issue 19 covered the model-supply-chain intrusions that put this on the agenda. What exists so far is a membership list, with no published standard.

[NVIDIA](#)

Bank of Baroda confirmed a breach traced to a compromised employee mailbox on 27 July. The 700GB-to-1TB volume in circulation is the extortion group's own claim and is independently unverified; the bank's confirmation covers the incident and its containment. [The Asian Banker](#)

Two more restructures on the same reading, 20–22 July. Monday.com cut a fifth of its staff — about 630 roles — while reaffirming full-year revenue growth of 19–20% and raising its margin outlook. Intel cut an undisclosed number from Data Center and AI, the group whose Q1 2026 revenue reached \$5.1bn, as part of a broader strategy to become, in Intel's words, "a more focused and efficient company". [TechCrunch](#), [Tom's Hardware](#)

OUTSIDE IN

The harness outlasts the model (2 min)

Zack Deveau

ZD

Application Security, Shopify

Shopify Engineering · 29 July 2026

[Read it →](#)

"The most durable advantage you can develop is a harness tuned to the particularities of your software development ecosystem." That is Zack Deveau, writing from Shopify's application security team on 29 July, at the end of roughly six weeks running an agent programme across the company's own code.

What was built

The agentic harness is the orchestration and verification layer a team builds around a model: the prompting, the partitioning of work into pieces a model can hold, the decisions about what it is shown, and the deterministic checks that judge what comes back. The orchestrator — the component that sequences those steps and decides what runs next — is code the team writes and owns. The model sits inside it as a replaceable part.

80+

Applications scanned

300+

Findings produced

2

Rated Critical

6

Weeks running

The cost envelope, stated out loud

What makes the post unusual is that Deveau publishes the running costs. "A full application scan with publicly available frontier models costs between \$50 and \$300", and "incremental diff scans are much cheaper, roughly \$5 to \$50." Over six weeks the system covered more than eighty unique applications and produced more than three hundred findings, two of which would have been rated Critical.

Those figures make the per-run cost legible. A team can answer what a second pass over the same codebase costs, what a nightly incremental sweep costs across an estate, and where the ceiling sits before anyone signs anything.

Where the durable investment sits

Model contracts get renewed, repriced and replaced. The partitioning logic, the verification steps, and the accumulated knowledge of how one particular codebase is shaped stay put across all of that. Deveau's argument is that the second category is where a team's engineering effort compounds, and it is the reason he frames the harness as the advantage worth building.

There is a staffing corollary. A harness is software: it has an owner, a test suite, a backlog and a maintenance cost, and none of that appears as a line item in a model contract. An agent programme run as a procurement exercise leaves nobody accountable for the layer that holds the institutional knowledge.

The takeaway: the durable asset in an agent programme is the orchestration and verification layer the team owns and can measure, and it is the part that survives a model swap.

[Read the full post: *shopify.engineering*](#) →

ON THE RADAR

Coming Up

AUG
EARLY



npm 2FA-bypass tokens lose sensitive-action capability — tokens that skip two-factor authentication can no longer perform sensitive account operations. Trusted publishing — releasing from CI on short-lived OIDC credentials instead of a stored token — is the documented migration path.



Swift CBPR+ starts rejecting unstructured postal addresses — under the cross-border payments and reporting rules that carry ISO 20022 into correspondent banking, an address must arrive with each element in its own field rather than as free text. Research commissioned by RedCompass Labs in March 2026 put nearly half of banks behind on readiness — a vendor-commissioned figure that pre-dates this window.



CMA evidence-gathering on Microsoft 365 concludes — the end of the evidence phase in the probe opened on 29 July. No finding has been made.



npm 2FA-bypass tokens lose direct publishing entirely — the second step of the same cutover, roughly five months after the first.



AWS ME-CENTRAL-1 and ME-SOUTH-1 remain open on the AWS Health Dashboard — AWS's most recent public update on both regions, dated 30 April 2026, said they had suffered damage as a result of the conflict in the Middle East and that restoration was expected to take several months. [AWS Health Dashboard](#)

THE LEADERSHIP READ

Napkin math as a leadership discipline

 (3 min)

Ten seconds. That was how long a search query took in the benchmark a product team used to choose one database over another, and it was the number that settled the decision. Simon Eskildsen, talking to Gergely Orosz on 21 July, worked out that the same query should have taken ten milliseconds — three terms matched across indexed documents, the resulting lists intersected, all of it at memory bandwidth across several cores. The two numbers were a thousandfold apart.

SE

Simon Eskildsen

Founder, turbopuffer; formerly Shopify
The Pragmatic Engineer · 21 July 2026

[Read it →](#)

The table behind the practice

Napkin math is Eskildsen's own term, and it is more literal than it sounds. Behind it sits an artefact: a table he maintains of roughly fifty first-principles numbers — memory bandwidth, storage cost per gigabyte, network and disk latencies — with a script that regenerates them so they stay current as hardware moves. The table exists to establish what a system ought to be capable of before any measurement of that system is trusted.

I hate benchmarks because making design decisions based on benchmarks is not a satisfying answer to me.

— Simon Eskildsen, The Pragmatic Engineer

Where the thousandfold went

The benchmark that produced the ten-second figure was running distributed queries across roughly a hundred nodes and reporting the result without accounting for p99 latency — the value below which ninety-nine of every hundred requests complete. The slowest node in each fan-out was setting the headline number. Eskildsen's conclusion was that the benchmark had measured the wrong thing, and that the arithmetic is what caught it.

The same arithmetic, applied to a bill

The second example is commercial. Cursor was spending around \$80,000 a month on indexing and search with another vendor. Eskildsen predicted from the same kind of calculation — work out what the resource consumption actually is, then add margin — that the same workload could run for a fraction of it. After moving to turbopuffer, the vector search database he founded, which keeps vectors cheaply in object storage and caches the hot set in memory, the bill came to roughly \$4,000 a month. The prediction became the product's differentiator, because the arithmetic said the headroom was there before any of it was built.

Why this is a leadership skill

Napkin math reads as an engineer's habit, and it is one. It is also a leadership one, because the practice does not require the ability to build the system. It requires a rough sense of what the hardware can do and the willingness to ask, out loud and in front of people who know more, why the observed number sits so far from it. A director who could not write the query can still ask what the physics allows. When the answer and the measurement are three orders of magnitude apart, the explanation is usually in how the measurement was taken — and that is a question a leader can put to a design review without being the expert in the room.

[Read the full article: newsletter.pragmaticengineer.com](https://newsletter.pragmaticengineer.com) →

This could be worth talking about: *when a vendor benchmark and a first-principles estimate disagree by three orders of magnitude, which one should a design review believe — and whose job is it to notice the gap?*

RECOMMENDED READING

This Week's Links

Early adoption of agentic coding tools by GitHub projects — Raida & Hou

[arxiv.org](#)

Blacksmith outage on 21 July 2026 — the full post-mortem

[blacksmith.sh](#)

GitHub Actions incident record, 19–20 July

[githubstatus.com](#)

Visa fiscal third quarter 2026 financial results

[investor.visa.com](#)

VOL. II

Engineering Digest

18–31 July 2026 · Weekly Edition · Issue #20

Next edition: 1–7 August 2026

Archive

#16 · Weekly

#17 · Weekly

#18 · Deep Dive

#19 · Weekly

#20 · Weekly